This document is used to explain all the different components of the GUI toolbox for running, testing, optimizing, and validating TI simulations.

Within this document, there will be explanations for each of the different pages, along with screenshots to try and make it as clear and user-friendly as possible.

Prerequisites

The document and most of what is present below builds upon the fact that the user is able to get the application up and running.

For that, there is the Github page! This is the link to the repository that is self-contained and also contains a README file for the GUI and another README for the entire repository. The readme file for the GUI contains almost everything you need to setup the pipeline and to be able to run it, and so this document is just a supplement with images to make it easier to setup and to run.

Most importantly to setup, you require a simnibs environment, and if you would like to run code on SCITAS, then a SCITAS account and ssh/login ready.

The simnibs environment can be found in the following locations:

Windows:

C:\Users\<username>\SimNIBS-4.xx\simnibs_env → refers to your distribution of simnibs

MacOS:

/Applications/SimNIBS-4.x/simnibs_env

Linux:

~/SimNIBS-4.x/

Once you know your simnibs environment path, you just need to clone it onto wherever you are setting up your repository (which is your target path)

If you have a conda environment - most straightforward, have not tested the other cases.

conda create -p <target_path> --clone <path_to_simnibs_env>
<target_path>/python.exe -m pip install dash plotly

Linux:

1. Copy the whole environment directory
cp -r /path_to_simnibs_env /target_path

2. Install packages using the copy's own python

/target_path/bin/python -m pip install dash plotly

Windows (not conda):

robocopy path_to_simnibs_env target_path /E

target_path\python.exe -m pip install dash plotly

For example, my target path was D:\envs\bids_ti_gui_env

Hence, you also require a python with version 3.11 (at the current state)

Running the GUI

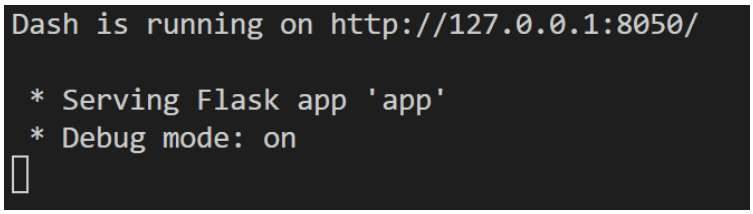
Assuming all the steps above ran with no issues and the paths are correct, you will just have to navigate to your directory (wherever you downloaded the repository) and then run the startup command from inside the repo.

<path_you_cloned_the_simnibs_environment_to>/python.exe code/gui/app.py

Given my case, where the target path was D:\envs\bids_ti_gui_env , I would type

D:\envs\bids_ti_gui_env\python.exe code\gui\app.py

Then, just open http://127.0.0.1:8050/. This should give you the running pipeline as a page.
When you run the command, you will also be able to click the link in the terminal:



As long as no other servers are using the 8050 port, you should be okay!

0) Setup

0.1) SCITAS Connection / Data storing and managing

There are several things the user must already have to be able to connect to SCITAS. (These are outside the GUI and if you've used SCITAS before at all these should probably already be there).

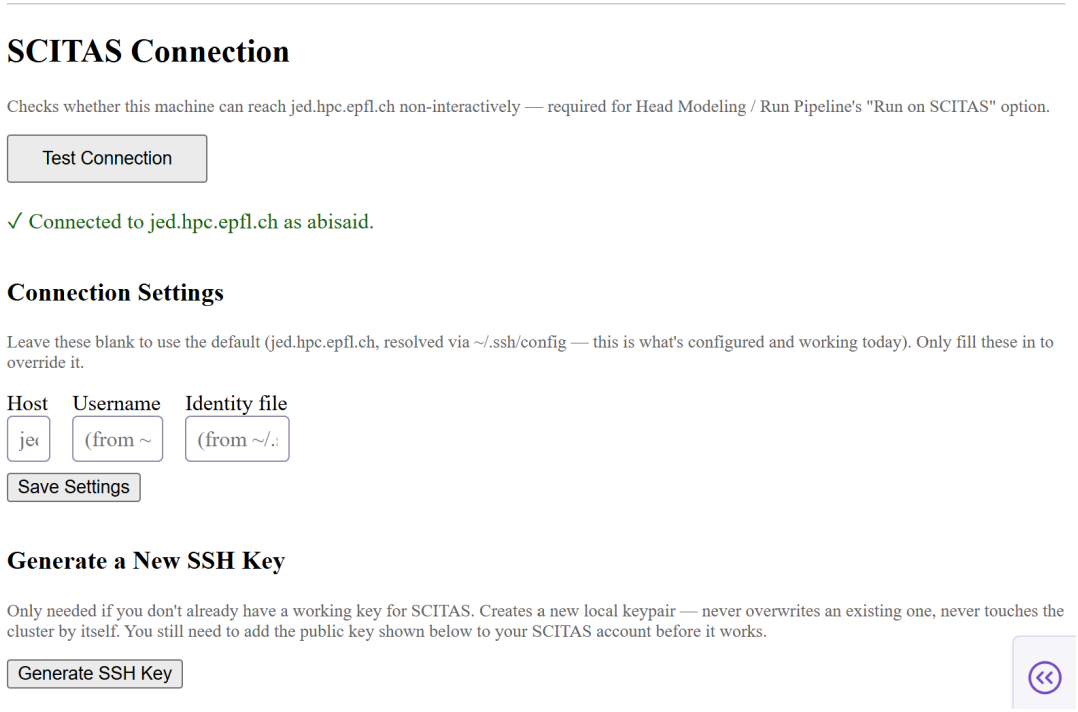
1) SCITAS user account (with a gaspar username)
2) ssh present on your computer (basically all computers come with this)
3) Connection to EPFL wifi (otherwise, use the EPFL VPN)

The above are all probably already there if you've used SCITAS at any point.
If you already have an SSH key, with your username and everything already existing, you should be able to press the Test Connection button and it should succeed.

Otherwise, you should add a key-pair using Generate a New SSH Key.

For the connection settings, they are already pre-built for EPFL.

1) Host: jed.hpc.epfl.ch
2) Username: usually taken from your own ~/.ssh folder, but you can set it with your gaspar username if need be.
3) Identity file: Only if you set up a new key using the Generate a New SSH Key button.



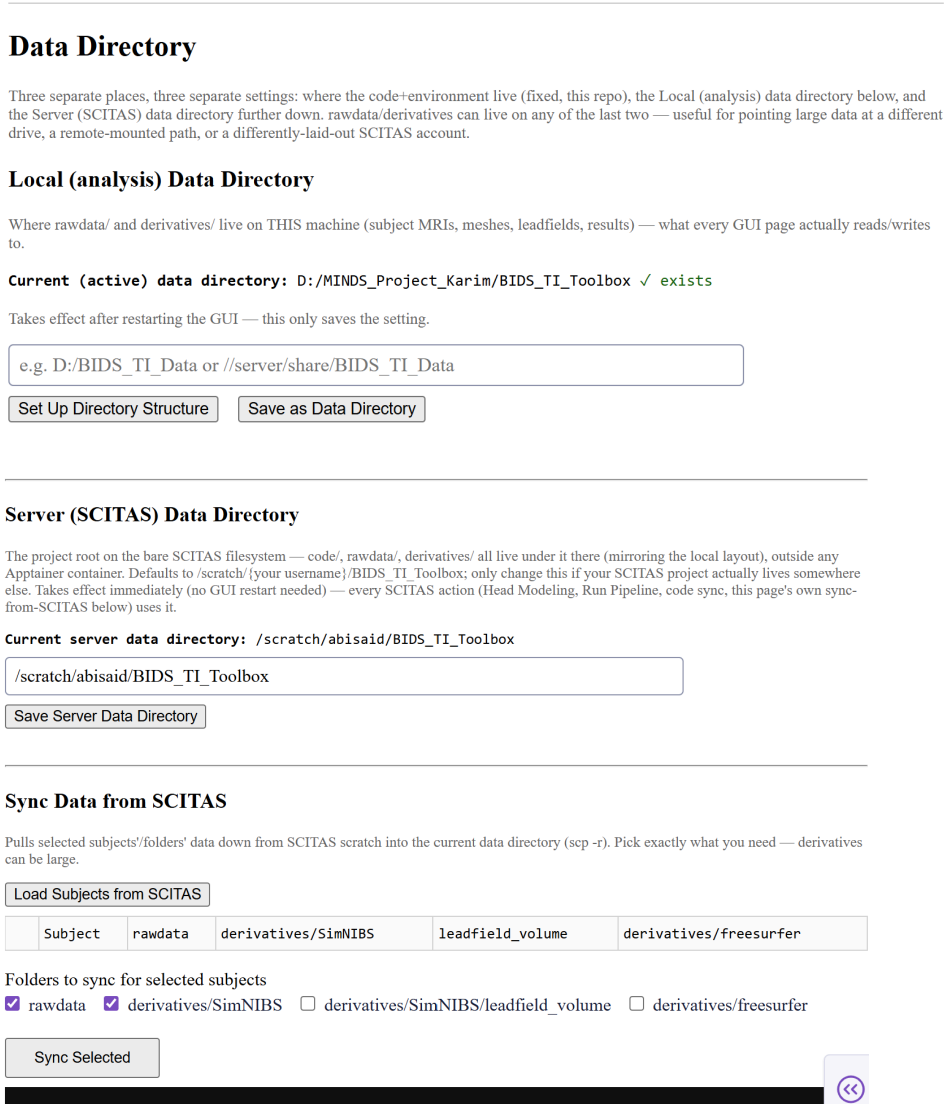
0.2) Data Directory

This is another prerequisite. This defines where you will have your data extracted, coded, and then placed.

It consists of your Local/Server Data Directory
You are able to connect it to your own directory if you wish to have the data locally, or to a folder on the \Hummel-Lab directory for example. Currently in the screenshot, I use my own hard drive (D:\), but any folder works. It also should contain the folders with your raw data (MRI scans, participant names, etc).

On SCITAS, you also have the directory where you will upload most temporary files and calculation-heavy things go. Since you might want to submit jobs to the cluster, then it would make sense to also program the /scratch/<username>/**BIDS_TI_Toolbox** as the folder where you'll put all your computations at. The folder must be named **BIDS_TI_Toolbox** since many of the scitas-calling code refer to this folder.

When you have such a folder, you will be able to sync data back to your local/server data directory (discussed in a later section).

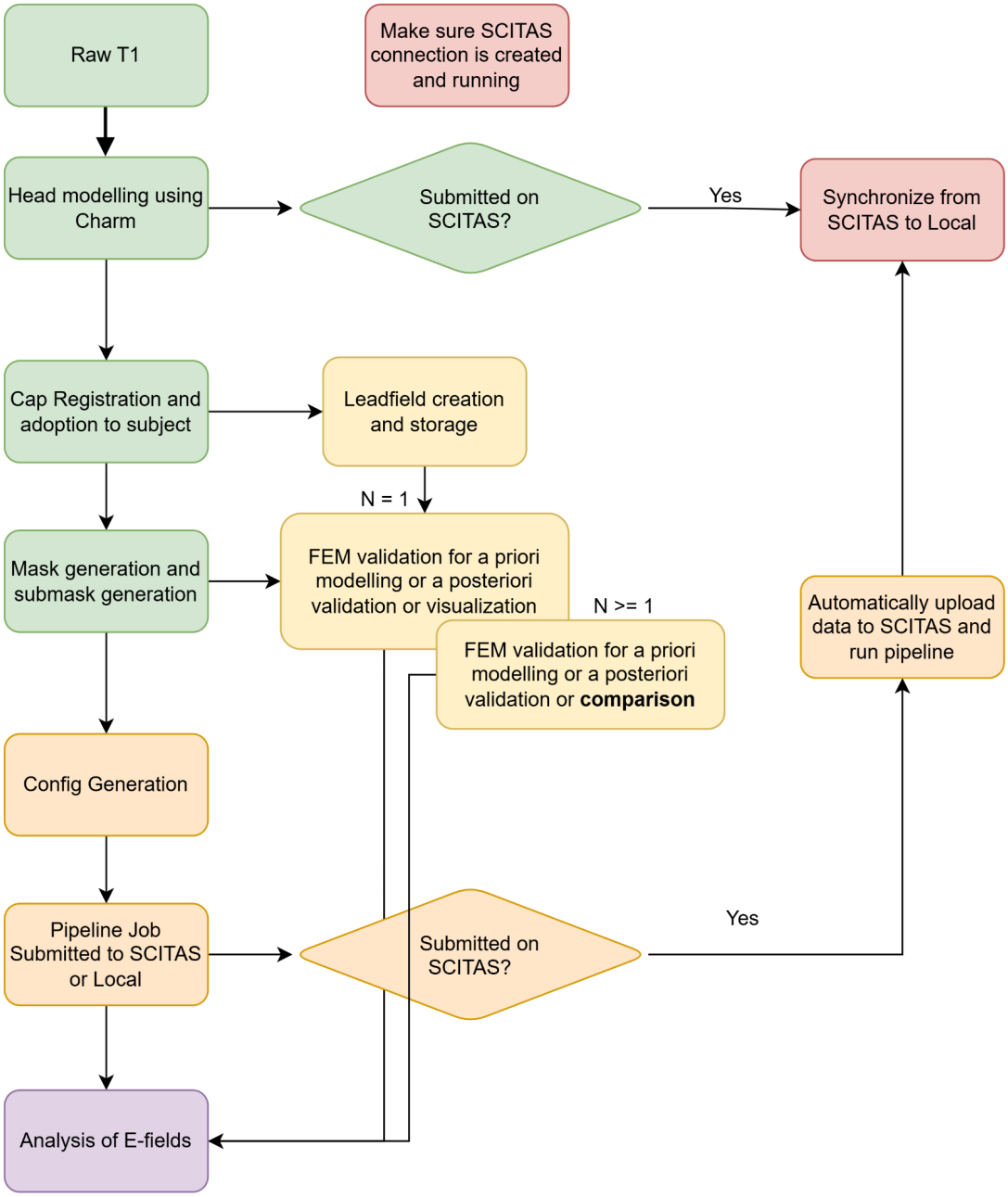


Finally, we also assume that the folder structure (which would be created using the “Set Up Directory Structure”) is unified across subjects and across users. To start, all you really need is one folder (your project directory), inside that folder is rawdata/sub-<id>/anat/T1w, or also the T2w scans.

```
<PROJECT_DIR>/
├── rawdata/sub-{id}/anat/
│   ├── sub-{id}_T1w.nii.gz            (required — Head Modeling / charm)
│   └── sub-{id}_T2w.nii.gz            (optional — improves charm segmentation quality)
└── derivatives/
    ├── SimNIBS/sub-{id}/               (generated by this toolkit — charm's m2m_{id}/,
    │                                     roi/ masks, eeg_positions/ registered caps,
    │                                     leadfield_volume/, comparison/ FEM+TI outputs)
    └── freesurfer/sub-{id}/           (optional — only if using the FreeSurfer atlas
                                          source, not yet usable per create_masks.py)
```

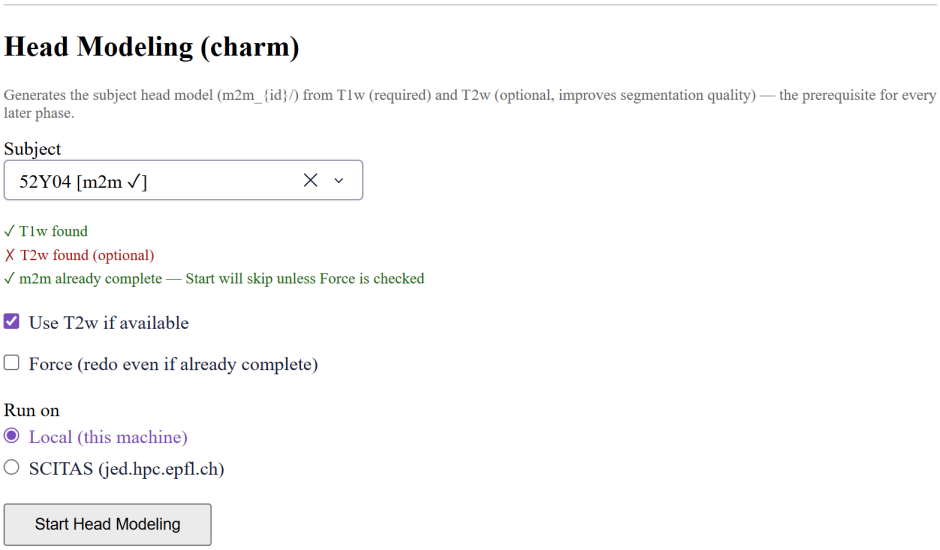
Using the GUI

In this section, we explain using multiple subsections how to utilize the GUI, the different inputs, and some examples.
Typically, given **only a raw T1 scan**, the pipeline would look as such:



1) Preprocessing:

1.1) Head Modeling:



We use this section to create a head model for any subject folder that exists inside your local directory, which again should fall under the pattern of local_directory\rawdata\sub-<id>\T1w.nii.gz

The toolbox automatically checks whether the m2m folder (which contains the outputs of SimNIBS charm) exists for a specific subject, and can hence optionally force the rerun or just skip it in case it exists already. It can also be submitted to scitas.

1.2) Mask Generation

Mask Generation

Subjects

52Y03 [m2m ✓, raw ✓], 73T02 [m2m ✓, raw ✗], 025 [m2m ✓, ra… 3 selected

Atlas Availability

Per subject, per atlas: ready to use, or what's missing (MNI warp field, subject labeling, atlas file, roi list, ...).

Subject	SimNIBS	BNA	Allen	FreeSurfer
52Y03	✓ ready	✓ ready	✓ ready	✗ not yet usable (see create_masks.py); aseg_path
73T02	✓ ready	✓ ready	✓ ready	✗ not yet usable (see create_masks.py); aseg_path
025	✓ ready	✓ ready	✓ ready	✗ not yet usable (see create_masks.py)

This section has multiple steps
Initially, you are allowed to select multiple subjects, and immediately what is available to you in terms of mask sources will be displayed. If you have the m2m file ready (charm ran successfully), then the SimNIBS mask will be available for use. The BNA and Allen masks are technically already the repository which you have already downloaded.

When you select the Mask Source, also immediately the necessary files will be told if they exist or not. If they do not exist for a specific subject, they should be either added or the subject will be unable to have a mask created.

Another source for the mask can also be an external mask. If you have the mask in some folder on your computer and is in MNI space (or subject space), you can select that mask and either register it to subject space if it isn't already registered, or adopt it into the subject space if it's already there.

New Mask — Source Configuration

Atlas source

Allen

Allen Brain Atlas (Ding 2020), ICBM 2009b Nonlinear Symmetric space.

Path check (per selected subject)

Subject	Check	Path
025	m2m folder	D:/MINDS_Project_Karim/BIDS_TI_Toolbox\derivatives\
025	labeling.nii.gz (Conform grid + SimNIBS labels)	D:/MINDS_Project_Karim/BIDS_TI_Toolbox\derivatives\
025	Conform2MNI_nonl.nii.gz (MNI warp field)	D:/MINDS_Project_Karim/BIDS_TI_Toolbox\derivatives\
025	atlas_path	D:\MINDS_Project_Karim\BIDS_TI_Toolbox\code\resourc
025	roi_list_path	D:\MINDS_Project_Karim\BIDS_TI_Toolbox\code\resourc
52Y02	m2m folder	D:/MINDS_Project_Karim/BIDS_TI_Toolbox\derivatives\
52Y02	labeling.nii.gz (Conform grid + SimNIBS labels)	D:/MINDS_Project_Karim/BIDS_TI_Toolbox\derivatives\
52Y02	Conform2MNI_nonl.nii.gz (MNI warp field)	D:/MINDS_Project_Karim/BIDS_TI_Toolbox\derivatives\
52Y02	atlas_path	D:\MINDS_Project_Karim\BIDS_TI_Toolbox\code\resourc
52Y02	roi_list_path	D:\MINDS_Project_Karim\BIDS_TI_Toolbox\code\resourc
73T01	m2m folder	D:/MINDS_Project_Karim/BIDS_TI_Toolbox\derivatives\
73T01	labeling.nii.gz (Conform grid + SimNIBS labels)	D:/MINDS_Project_Karim/BIDS_TI_Toolbox\derivatives\
73T01	Conform2MNI_nonl.nii.gz (MNI warp field)	D:/MINDS_Project_Karim/BIDS_TI_Toolbox\derivatives\
73T01	atlas_path	D:\MINDS_Project_Karim\BIDS_TI_Toolbox\code\resourc
73T01	roi_list_path	D:\MINDS_Project_Karim\BIDS_TI_Toolbox\code\

New Mask — Source Configuration

Atlas source

External mask — type a path to any NIfTI mask

Any standalone NIfTI mask, warped into each selected subject through the same per-subject MNI warp every atlas above uses — no registry entry needed. Only accurate if the file's own space already agrees with charm's MNI target (same template SimNIBS itself registers to); a template this project hasn't used before needs a one-time correction first, not something this page does for you. The name below is used exactly as typed — no atlas suffix is added.

Path check (per selected subject)

Subject	Check	Path	Status

Region selection

Path to mask file (.nii / .nii.gz)

e.g. D:\project\external_masks\hippocampus.nii.gz

This mask is already in:

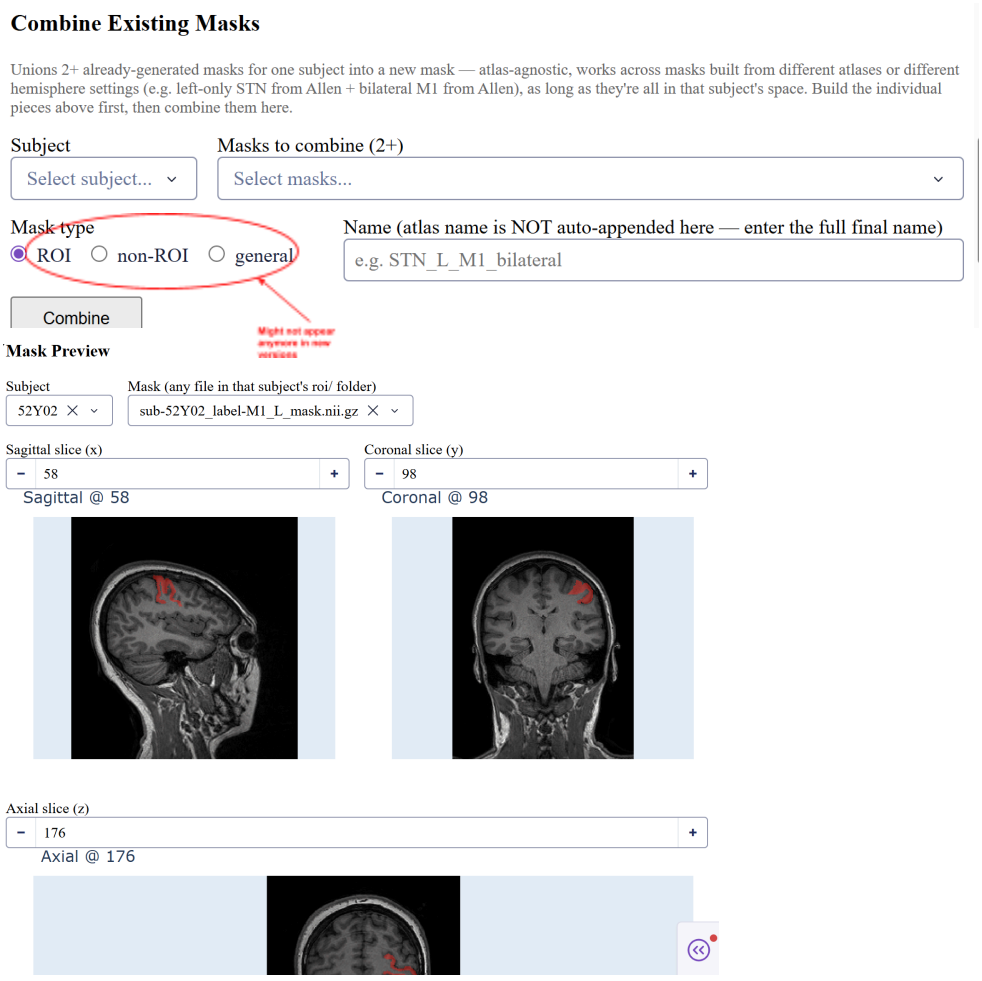
- (●) MNI-ish space — needs warping into each selected subject, same as an atlas region
- ( ) One specific subject's own native space already — no warping, just placed into that subject's roi/ folder (resampled onto the reference grid if needed, but no MNI warp). Only usable for exactly one subject — select just that one above.

After checking all the necessary files for the corresponding path exist, then move on to the region selection step. In this step, you either select the regions (one or many) that you wish to have the mask, based on the previously selected source, and then you choose whatever configuration you need (left, right, both hemispheres), and also input a name (in this example it is M1), which will be appended to the name of the mask later on used for searching. As you can see in this example, 52Y02 already has a mask with that name existing, and so we must either un-select it, rename the mask, or force the rewrite.

You may also combine existing masks for the same subject into a single unified one (useful when there are many separate ROIs.

Finally, there is a visualization function that allows you to see the mask that you have generated and be able to validate it. (In the example, again we show the left M1).

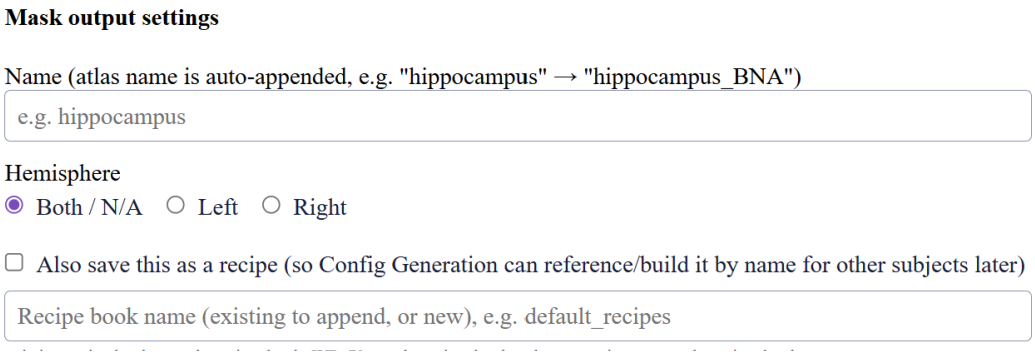
You can also export the mask as a mesh file which allows you to view the ROIs in .gmsh.



1.2-a) Mask recipe book

Now, if you do not wish to create the masks every single time for every single subject, the current update of the system has a “recipe book”. With this, any mask you create can be added as a recipe which you can apply for any other subject. It will be placed within code\gui\mask_recipes. Placing the same recipe book name for multiple masks will append the new masks into the recipe book.

This is available for both the single mask generation as well as for the mask combination.



1.3) Cap Registration

Since we would like the optimization procedure to be able to accept any kind of eeg cap (especially since we might either want the cap to be obtained through neuronavigation, or run on heavier caps such as 128 electrodes, or lighter like a 32 electrode cap), then we have multiple caps that are generated by SimNIBS already, along with an included BioSemi32 cap. These are all in MNI space, and so must be placed into subject space through registration, by performing the same warp that SimNIBS performs on a subject. If the ras file exists, then we can simply “adopt” the cap EEG locations to the subject, without performing the registration. There is also a function to help view the electrode positions on any registered cap.

Cap Registration

Subjects

52Y09 [m2m ✓, raw ✓]

Cap (project + SimNIBS built-in)

EEG10-10_UI_Jurak_2007 (simnibs)

...or a custom path (overrides the dropdown if set)

e.g. D:/path/to/my_cap.csv

✓ 80 electrodes — D:\envs\bids_ti_gui_env\Lib\site-packages\simnibs\resources\ElectrodeCaps_MNI\EEG10-10_UI_Jurak_2007.csv

Custom path space (only matters when a custom path is set — the dropdown caps above are always MNI space)

- (●) MNI space (needs warping — default)
- ( ) Already in subject space (e.g. digitized positions, or a previously-registered file) — no warping, one target subject

Registration Status

Subject	Ready (m2m + mesh)	Already registered	Registered path
52Y09	✓ ready	✓ yes	D:/MINDS_Project_Karim/BIDS_TI_Toolbox\derivatives\SimNIBS\

Register

Select All Ready

	Subject	Output path
☑	52Y09	D:/MINDS_Project_Karim/BIDS_TI_Toolbox\derivatives\SimNIBS\sub-52Y09\m2m_52Y09\eeg_positions\E

⚠ 1 cap(s) already registered and will be overwritten: 52Y09

Cannot register yet — confirm overwrite to proceed

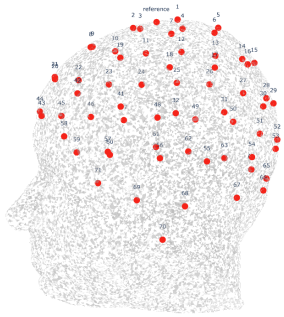
Register

Cap Preview

Decimated scalp surface + registered electrode positions for one subject — a quick sanity check, not a substitute for the .msh export.

Subject: 73T03

Registered cap: easycap_BC_TMS64_X21



2) Simulation

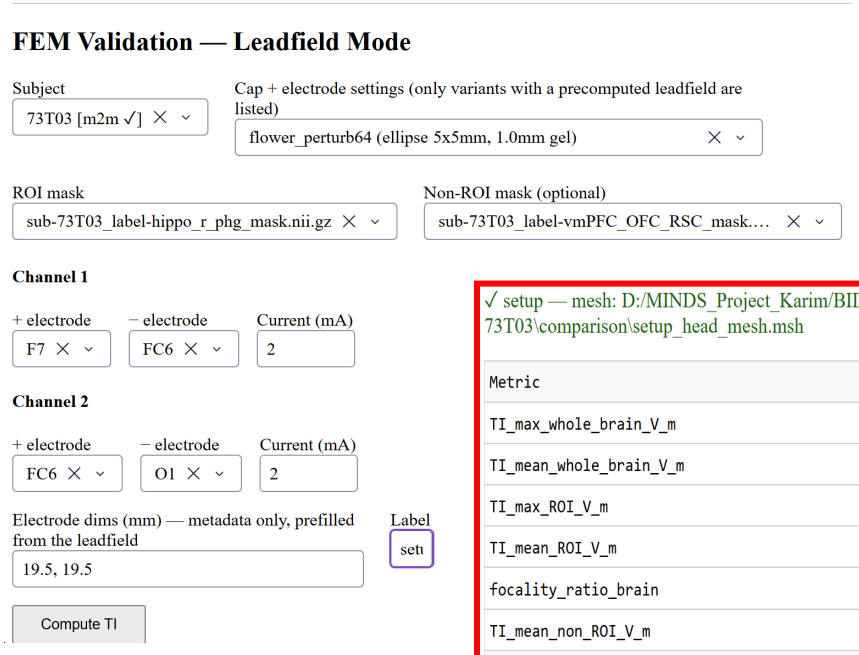
2.1) FEM Validation

This section is used to calculate the values of TI fields inside the regions and masks built before. It can be used to manually test positions or run a quick sanity-check on what the amplitudes for a certain subject might look like.

Ideally, you would want to have a leadfield present for the certain cap that you will be working with. However if not, there are multiple other options that can be easily employed.

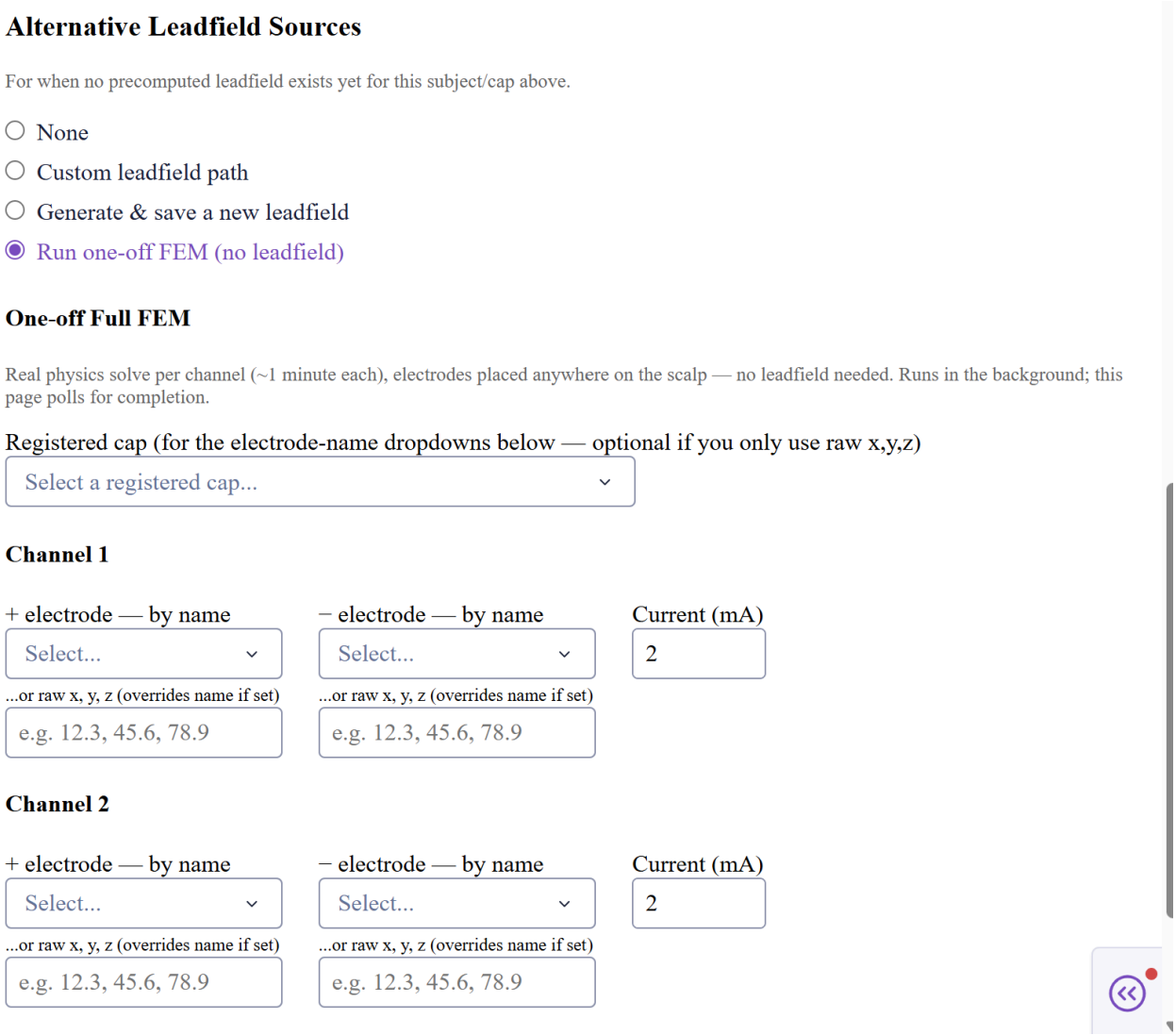
1) Generate the leadfield live: This takes some time, but it only needs to be done once and gets saved to your directory anyway.
2) Run FEM once: If you don't have space or don't want to generate a leadfield, you are also able to run the FEM directly to obtain results.
3) If you have a leadfield saved somewhere differently, you can just input the path of that leadfield and the electrode positions.

The example is shown below.



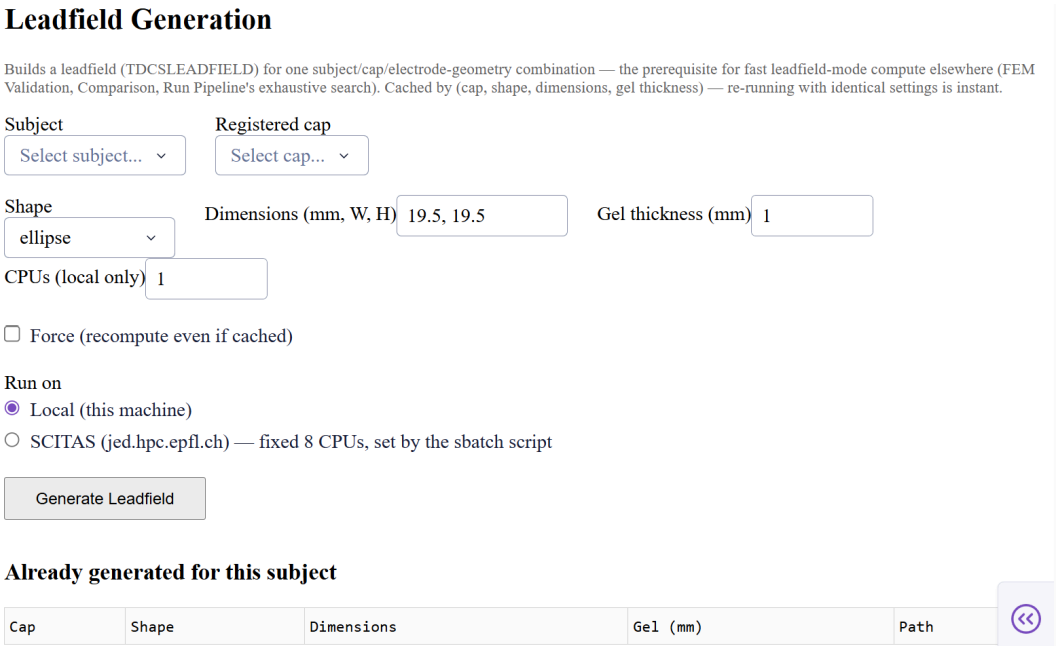
✓ setup — mesh: D:/MINDS_Project_Karim/BIDS_TI_Toolbox\derivatives\SimNIBS\sub-73T03\comparison\setup_head_mesh.msh

Metric	Value
TI_max_whole_brain_V_m	3.4475
TI_mean_whole_brain_V_m	0.3046
TI_max_ROI_V_m	1.6227
TI_mean_ROI_V_m	0.4280
focality_ratio_brain	1.4052
TI_mean_non_ROI_V_m	0.2844
focality_ratio_non_roi	1.5049
TI_mean_WM_V_m	0.3738
TI_max_WM_V_m	3.4475
TI_mean_GM_V_m	0.2471
TI_max_GM_V_m	1.3269



2.2) Leadfield Generation

In the case that you would like to run validation multiple times, without having to run the singular FEM runs every time, you could run leadfield generation one time for a subject, and then it would be saved and allow you to easily run multiple different runs for different locations quickly. The cost is effectively a singular leadfield load, then the computations are incredibly fast (pure algebraic).



2.3) Comparison

Since it would also be interesting to compare two (or more) subjects/montages/caps… it would be useful to have a page that is effectively the same as the above FEM validation page, but be able to run and compare multiple simulations side-by side.

We can compare across

1) Different ROIs
2) Different non-ROIs
3) Different Subjects
4) Different caps
5) Different montages

An example of the page is shown below:

Comparison

Run multiple FEM/TI setups side by side. Rows can share a subject (different montages), share a montage (different subjects), or be fully independent.

Cap

☑ Use the same cap for all setups

Common cap

BioSemi32_MNE (project)

73T01: ✓ registered leadfield ✓ (leadfield mode ready)

73T02: ✓ registered leadfield ✓ (leadfield mode ready)

Electrode Settings (leadfield rows only)

Which cached leadfield variant to use for a given cap — a cap can have more than one, computed with different electrode dimensions/gel thickness.

☑ Use the same electrode settings for all setups

Dimensions (mm) 19.5 19.5 Gel thickness (mm) 1

As you can see, the naming of the ROI when generating it is important, since this allows us to select the same ROI for multiple subjects. (otherwise, if we want to compare different ROI masks for example (let's say the BNA vs the Allen atlas on the right hippocampus), then simply un-select the "Use the same ROI for all setups", and another column will appear in the table before you click on run.

ROI

☑ Use the same ROI for all setups

Atlas Region

Select atlas... Select region...

...or pick from mask names that already exist (checked across the table's current subjects)

Select existing mask name...

Search

OFC_BNA (2/2 subjects)

RSC_BNA (2/2 subjects)

hippo_r_phg (2/2 subjects)

right_hc_phg_hierarchical_BNA (2/2 subjects)

multi-region union's custom name)

...or pick from mask names that already exist (checked across the table's current subjects)

Select existing mask name...

Resolved non-ROI label (auto-filled above — or type your own)

Setups

Mode: "leadfield" (fast — Ch1/Ch2 must be electrode names) or "oneoff" (real FEM solve — Ch1/Ch2 accept a name or raw "x, y, z"). Label names the output files and the comparison column below; auto-filled if left blank.

Add rows for subject(s)

73T01, 73T02 2 selected

Add Rows for Selected Subjects

	Subject	Mode	Ch1+	Ch1-	Ch1 mA	Ch2+	Ch2-	Ch2 mA	Label
×	73T01	leadfield			2			2	
×	73T02	leadfield			2			2	

Add Empty Row

Run Comparison

Setups

Mode: "leadfield" (fast — Ch1/Ch2 must be electrode names) or "oneoff" (real FEM solve — Ch1/Ch2 accept a name or raw "x, y, z"). Label names the output files and the comparison column below; auto-filled if left blank.

Add rows for subject(s)

73T01, 73T02 2 selected

Add Rows for Selected Subjects

	Subject	Mode	Ch1+	Ch1-	Ch1 mA	Ch2+	Ch2-	Ch2 mA	ROI mask	Non-ROI mask	Label
×	73T01	leadfield			2			2	OFC_BNA		
×	73T02	leadfield			2			2	hippo_r_phg		

Add Empty Row

Run Comparison

Export filename Export Results (CSV)

CSV export is also possible.

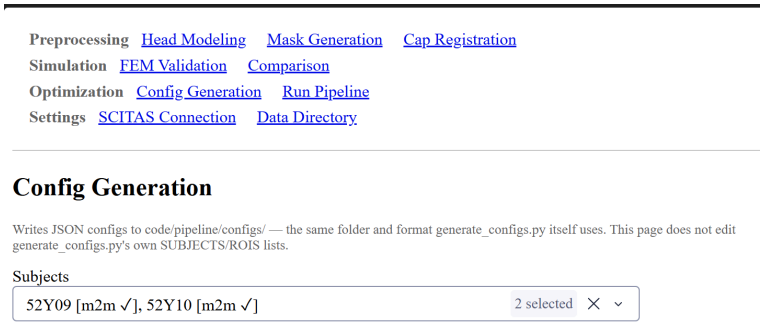
3) Optimization

This is the most important part of the GUI, since it allows us to run the optimization procedure created in the pipeline part of the code. It could be updated in the future to access multiple optimization algorithms if needed, but that would have to be extended later on. As for now, the functionality relies on a single optimization pipeline, which is the hierarchical search of the pipeline.

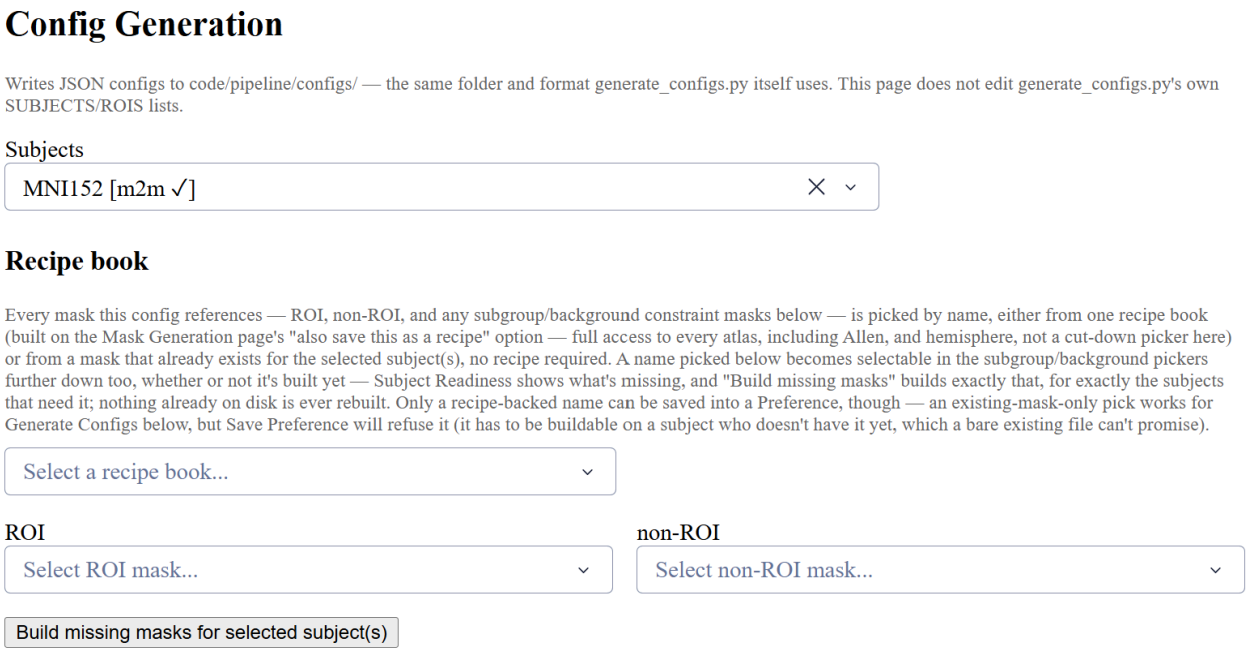
3.1) Config Generation

Every run of the pipeline requires a separate configuration file, created in this page. Within the page, you will be able to set all the different parameters:

1) Subject: It's possible to select single or multiple subjects, but if it is multiple subjects they should all have the registered cap that you wish to use and the ROIs with common naming as well.



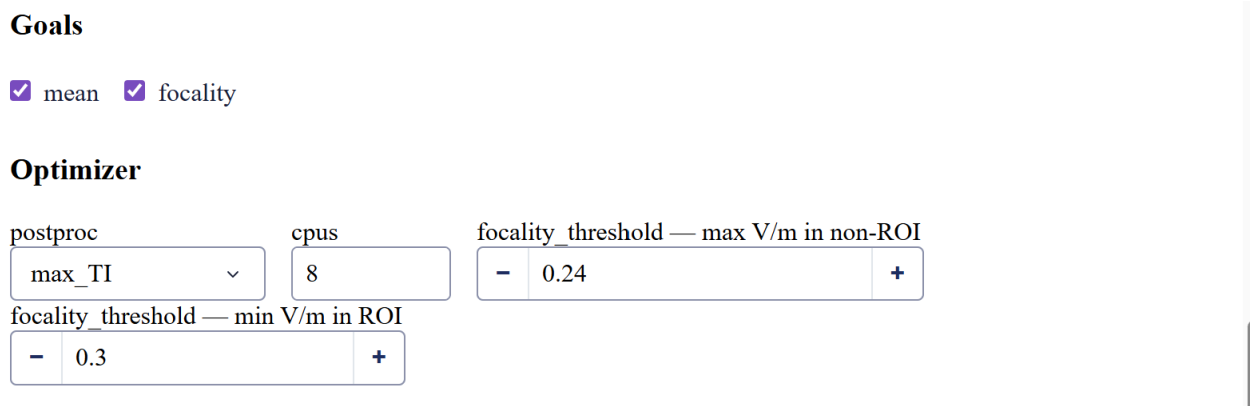
2) ROI and non-ROIs. You can select the specific ROIs and nonROIs you want to use, in the case that you have them loaded for one or more subjects already, you just select them in the ROI mask. However, if they are not generated before they can be generated instantly given they already have a recipe book. If they do not, the generation skips it and the pipeline will fail. Furthermore, if you select an ROI that is existing for one subject, but not for another, and not in the recipe book, it will also fail. However, a warning sign will show, so you can prepare and create the mask beforehand.



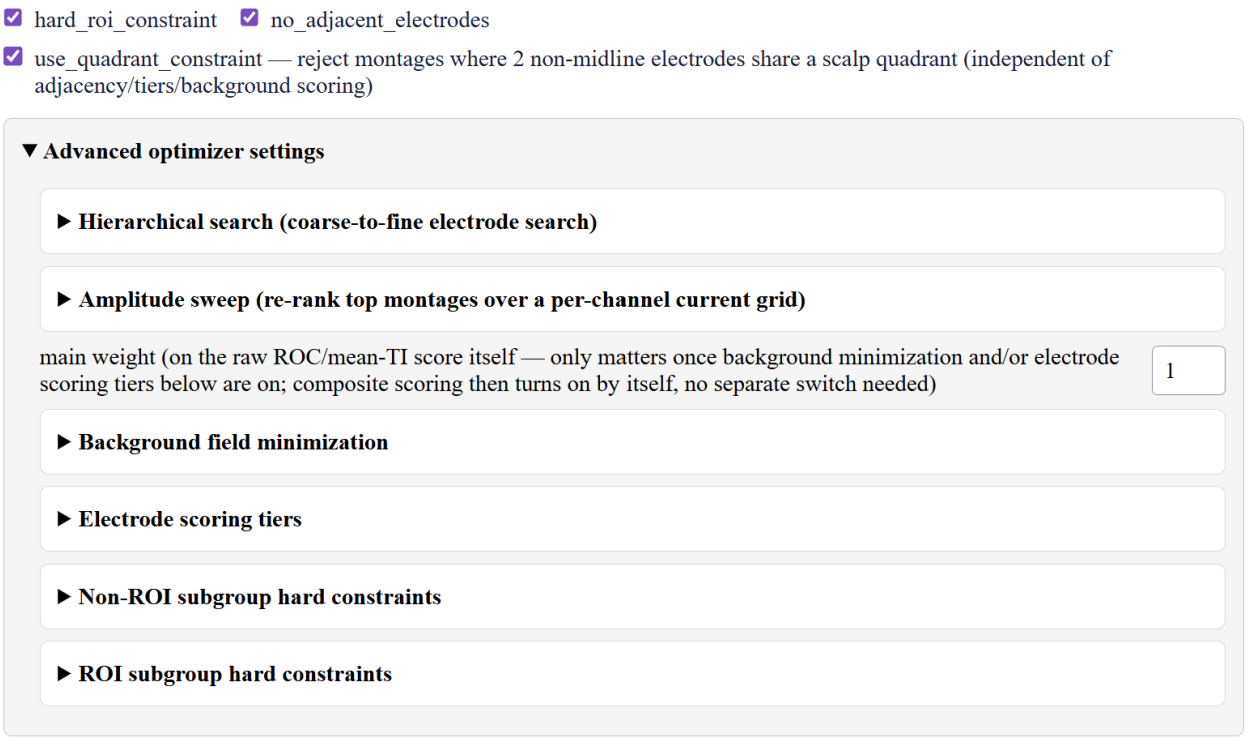
3) EEG Cap: If you wish to create multiple configs then you must make sure that all the subjects have the existing cap

4) Goals: Can be one of Mean/Focality or both (creates 2 configs)

5) Optimization fields: postprocessing and cpus are SimNIBS and SCITAS-specific parameters, no need to change them.

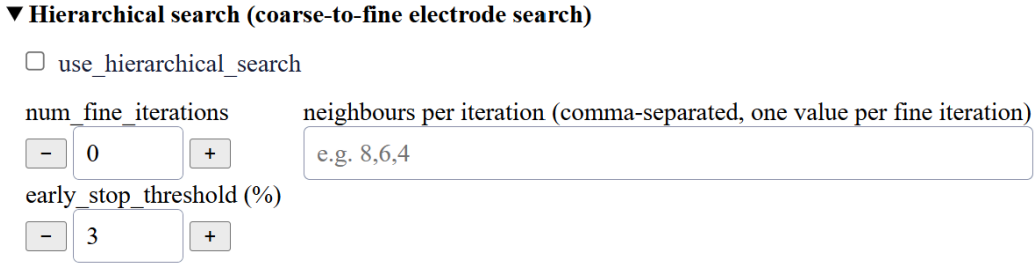


6) **Advanced settings:** These are the extra settings you can select and add to make the optimization really project-specific:

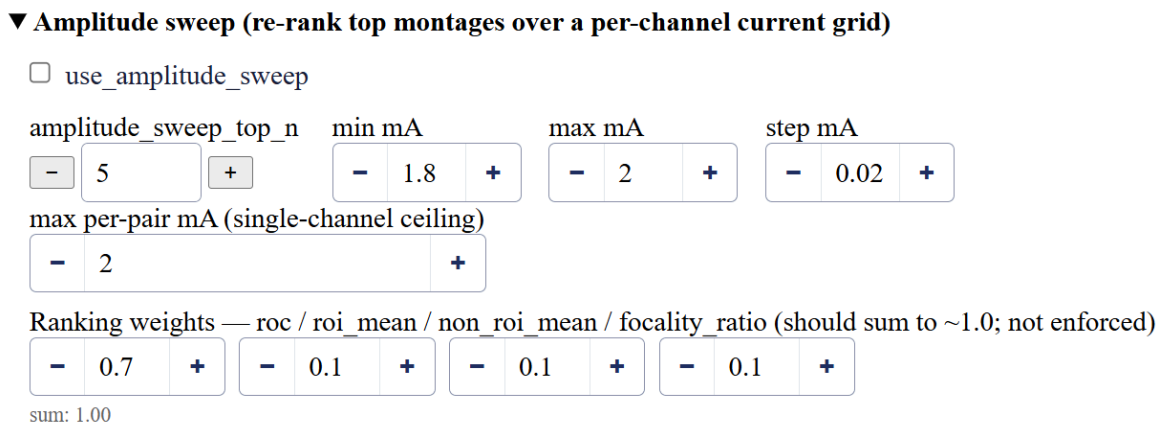


a) Hard roi constraint: the ROI may not decrease below a certain level.
b) No adjacent electrodes: we take any electrodes which are direct neighbours of the current electrode (anything less than 1.95x the nearest electrode neighbour) and we exclude them from being available for analysis, so that no electrode can have its neighbor also stimulating (for source isolation and crosstalk)
c) Quadrant constraint: Every electrode has to lie in a different quadrant of the brain, but the centerline is available for all electrodes.

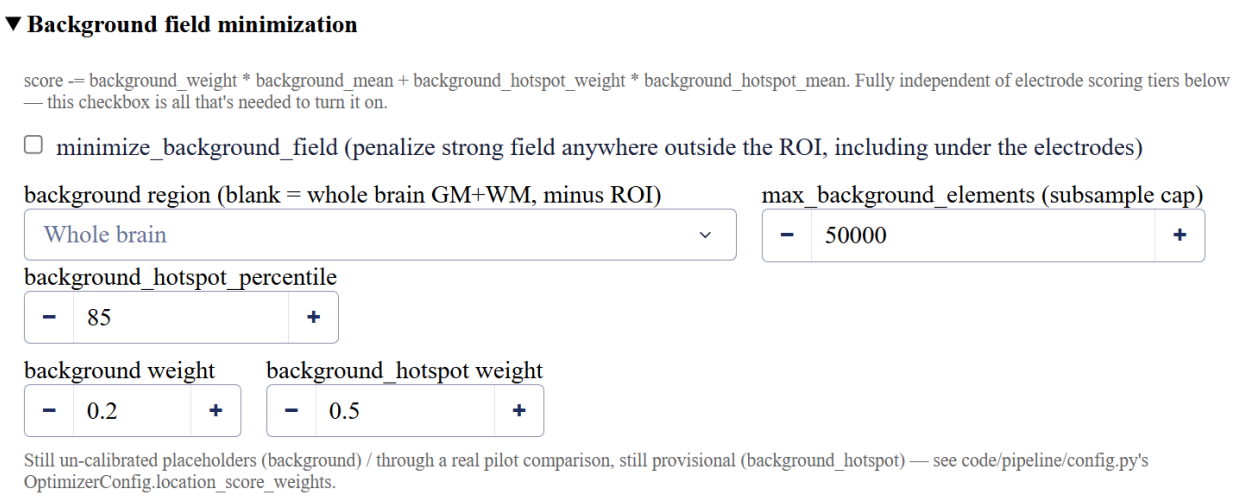
d) Hierarchical search: This is to be able to search more electrodes multiple times, without having the pipeline time balloon as we add more and more electrodes. With this, the time increases linearly while the number of electrodes can also increase, instead of having the time increase exponentially. (Personally, I do 1 fine iteration with 8 neighbors). The early stop threshold is the percent increase in score before it's considered to be the same.



e) Amplitude Sweep: This is to check whether we can use different amplitude ratios than 2:2, and checks (after the location has been specified) if we can achieve better focality with a different amplitude ratio.

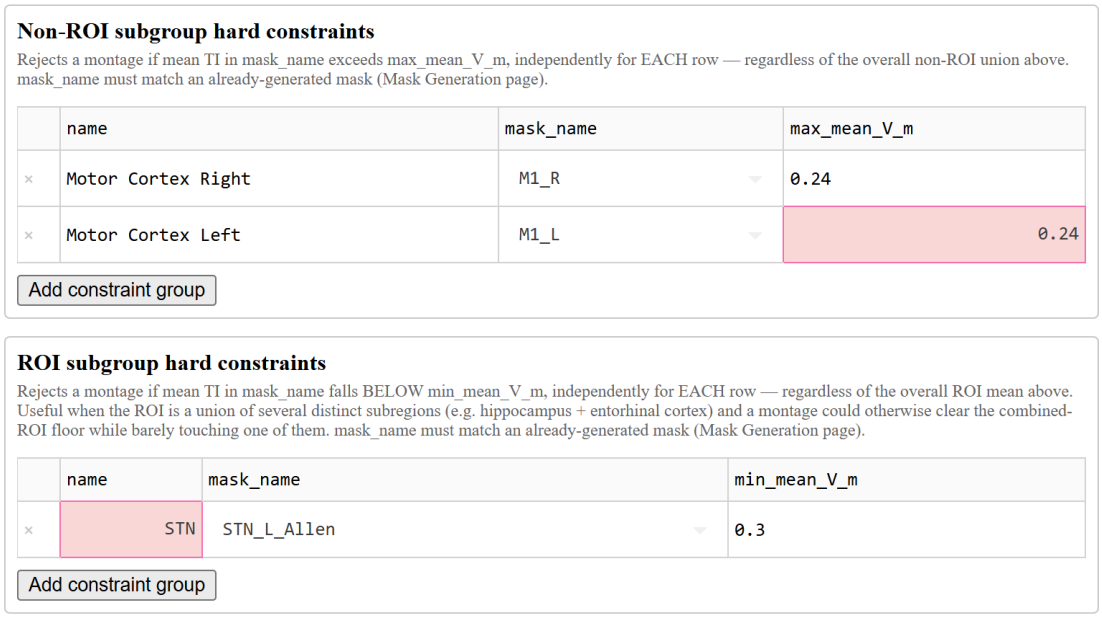


f) Background field minimization: adds another scoring condition, to lower the field in the nth percentile of the brain. 85th percentile means that the highest 85% of the brain's TI field (excluding the ROI) should be minimized. This allows us to push backwards the higher tail of the whole-brain group.



g) ROI and non-ROI subgroups: In this case, we can divide the ROI and non-ROI into different subgroups. This would be useful especially in cases where one group (ROI for instance) is unilateral, and the other group is bilateral.
Take the example of the left STN as the ROI and the M1 as the non-ROI. Suppose the groups are not divided, and the constraint is at 0.24V/m for the non-ROI. Given that the left STN is unilateral, the electrodes might be shifted more as a group towards the left, causing the TI field to be (for example) 0.26 in the left M1, and 0.21 in the right M1, averaging to < 0.24 V/m overall, but still having the mean

ROI inside the left M1 being above the threshold. So we can introduce as many non-ROI subgroups as we wish.



Non-ROI subgroup hard constraints

Rejects a montage if mean TI in mask_name exceeds max_mean_V_m, independently for EACH row — regardless of the overall non-ROI union above. mask_name must match an already-generated mask (Mask Generation page).

	name	mask_name	max_mean_V_m
×	Motor Cortex Right	M1_R	0.24
×	Motor Cortex Left	M1_L	0.24

Add constraint group

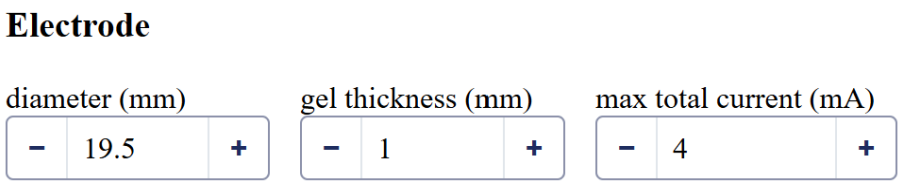
ROI subgroup hard constraints

Rejects a montage if mean TI in mask_name falls BELOW min_mean_V_m, independently for EACH row — regardless of the overall ROI mean above. Useful when the ROI is a union of several distinct subregions (e.g. hippocampus + entorhinal cortex) and a montage could otherwise clear the combined-ROI floor while barely touching one of them. mask_name must match an already-generated mask (Mask Generation page).

	name	mask_name	min_mean_V_m
×	STN	STN_L_Allen	0.3

Add constraint group

7) Electrode parameters: Self explanatory, size and thickness



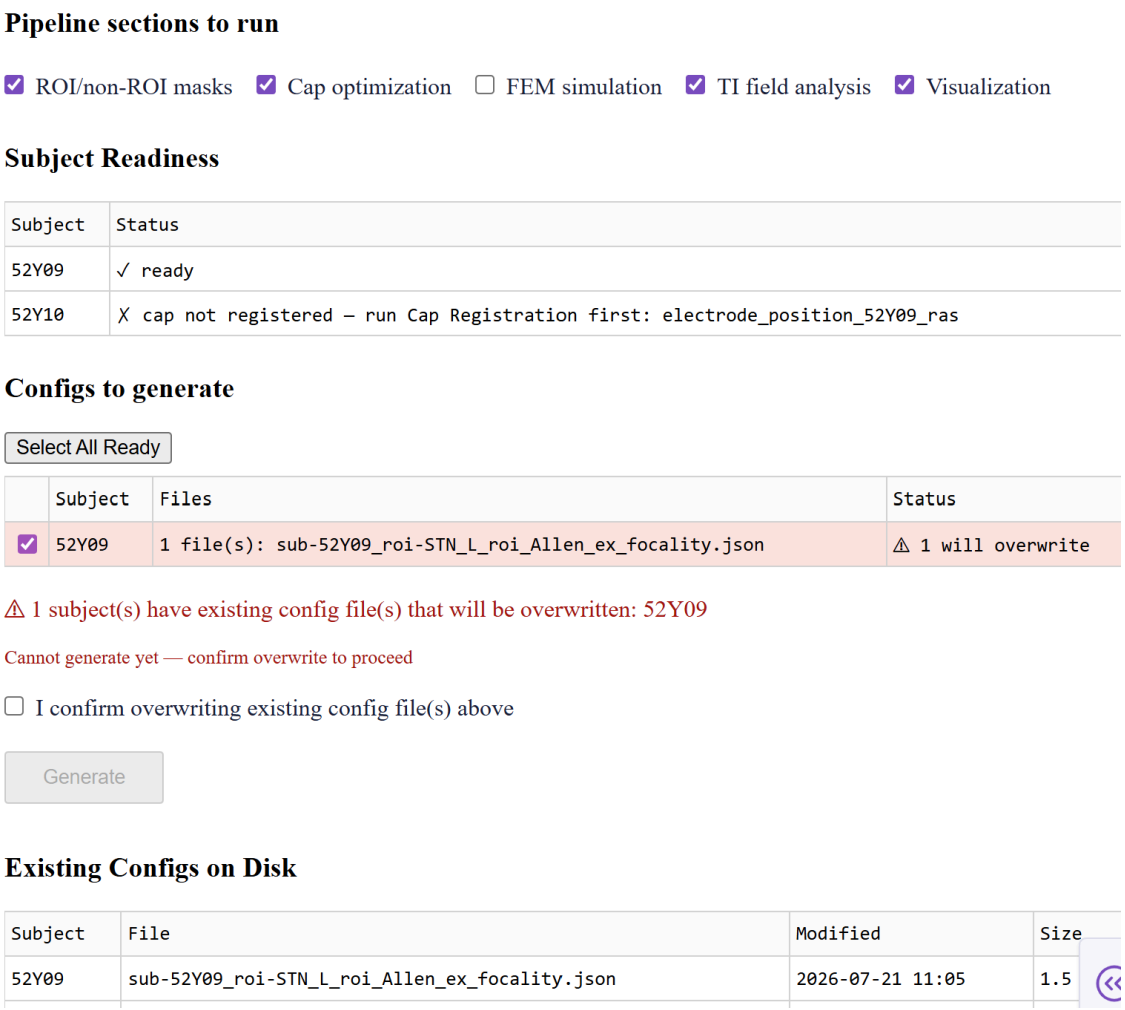
Electrode

diameter (mm) − 19.5 +

gel thickness (mm) − 1 +

max total current (mA) − 4 +

8) Pipeline sections: Since we might want to only run leadfield generation for example, we added the possibility to run only singular parts of the optimization pipeline.



Once generated, the config files are what will be submitted and contain all the information needed to run the pipeline. Note that the name of the json also has most of the information required.

3.1-a) Saving Config preferences

Similarly to the mask recipe book, we can also save a config preference. This will allow us to basically run any config we like at any time we like for new subjects, without having to recreate the masks or the config. Basically becomes an automated one-click config generation that runs through the entire pipeline until the generation.

Preference

Save everything above (cap, every mask — ROI, non-ROI, and any additional ones — every optimizer/electrode/flag setting) as a named, reusable Preference — not tied to any subject. Apply it to a new subject from the Apply Preference page: it builds the masks, registers the cap, and generates the config, so you don't have to set all of this up again every time.

Preference name, e.g. hippocampus_focality

Save as Preference

☐ overwrite the existing preference with this name

3.2) Apply Preferences

Given that you created a preference of a config, you can apply it at any time to any subject, create the masks, and run through the setting as mentioned above.

Apply Preference

Runs a saved Config Generation Preference against one or more subjects: head modeling (optional, local only — see the toggle below) → ROI/non-ROI/extra masks → cap registration → config generation, in that order, stopping a subject's own chain at the first failing step. Never submits the actual optimization/simulation job — that stays a separate, manual step on Run Pipeline once the configs exist.

Preference

WP_73_config_setup

- Cap: EEG10-10_Neuroelectrics.csv
- ROI: ROI_HC_EC (combine)
- non-ROI: prefrontal_nonROI (combine)
- Other masks (constraints/background): EC_BNA, HC_BNA, ofc_BNA, rsc_BNA, vmpfc_BNA
- Goals: focality

Subjects

MNI152

☐ Run charm locally for any subject whose m2m doesn't exist yet (otherwise that subject's chain stops there with a message instead of running a multi-minute operation unannounced)

Preview

Green = already done. Yellow = not done yet, Apply will do it. Red = blocks the chain for that subject until you address it.

Subject	Step	Status
MNI152	charm	✓ already done
MNI152	mask: EC_BNA	will be built
MNI152	mask: HC_BNA	will be built
MNI152	mask: ROI_HC_EC	will be built
MNI152	mask: ofc_BNA	will be built
MNI152	mask: prefrontal_nonROI	will be built
MNI152	mask: rsc_BNA	will be built
MNI152	mask: vmpfc_BNA	will be built
MNI152	cap registration	✓ already registered

3.3) Submitting Configs

Finally, you will submit the config files. However, note that when running on SCITAS, there happens to be an issue when submitting many configs at the same time, due to having to upload many files (m2m, leadfield if exists, mesh, etc) which will cause many back and forth calls to the server and have the system crash. It's recommended to submit one job at a time, make sure it's running (using squeue on SCITAS), and then submit another. If the optimization finishes on time, it will attempt to delete temporary files and upload the results on the server. To copy them locally, refer to the Data Directory section.

Run Pipeline

Runs already-generated config(s) from code/pipeline/configs/ (see Config Generation) through run_pipeline.py. Locally, project_dir/cap_csv/bna_atlas_path are overridden to this machine's paths — the config file on disk is not modified.

☐ Force re-run of every section (even if output already exists)

Run on

○ Local (this machine)

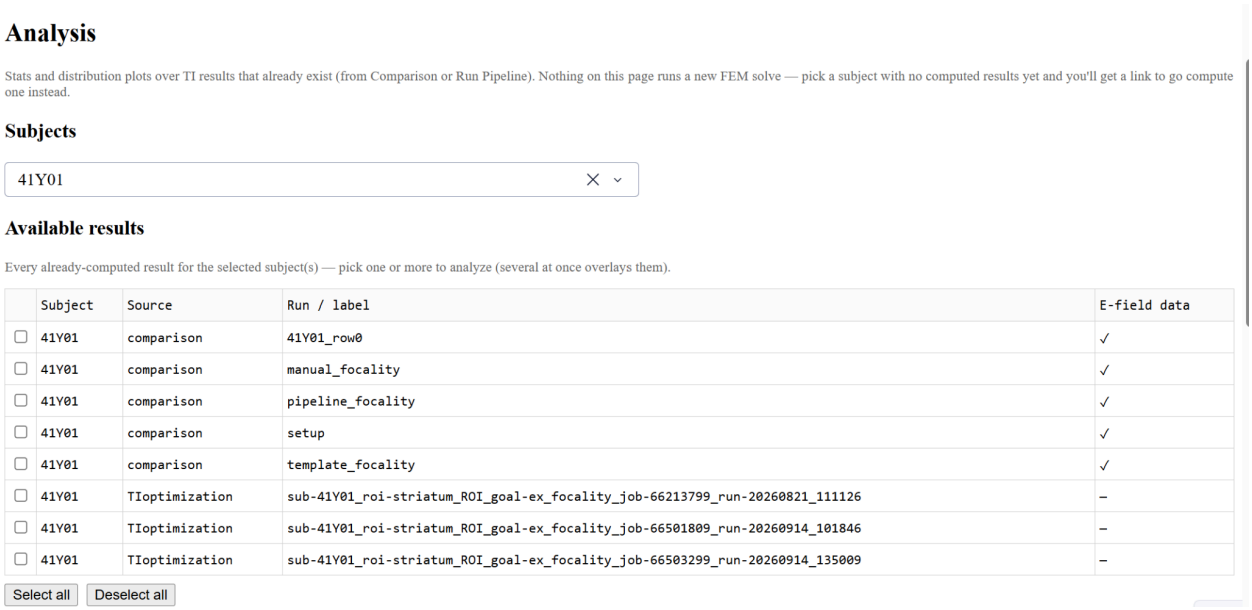
◉ SCITAS (jed.hpc.epfl.ch)

Available Configs

Select one or more configs — each starts as its own independent job (local: one background thread each; SCITAS: one sbatch submission each, run concurrently on the cluster) using the Force/Run-on settings above.

	Subject	ROI	Goal	File
☐	52Y02	STN_L_roi_Allen	focality	sub-52Y02_roi-STN_L_roi_Allen_ex_focality.json
☐	52Y03	STN_L_roi_Allen	focality	sub-52Y03_roi-STN_L_roi_Allen_ex_focality.json
☐	52Y03	STN_L_roi_Allen	mean	sub-52Y03_roi-STN_L_roi_Allen_ex_mean.json
☐	52Y04	STN_L_roi_Allen	focality	sub-52Y04_roi-STN_L_roi_Allen_ex_focality.json
☐	52Y04	STN_L_roi_Allen	mean	sub-52Y04_roi-STN_L_roi_Allen_ex_mean.json
☐	52Y05	STN_L_roi_Allen	focality	sub-52Y05_roi-STN_L_roi_Allen_ex_focality.json
☐	52Y05	STN_L_roi_Allen	mean	sub-52Y05_roi-STN_L_roi_Allen_ex_mean.json

4) Analysis



Analysis

Stats and distribution plots over TI results that already exist (from Comparison or Run Pipeline). Nothing on this page runs a new FEM solve — pick a subject with no computed results yet and you'll get a link to go compute one instead.

Subjects

41Y01

Available results

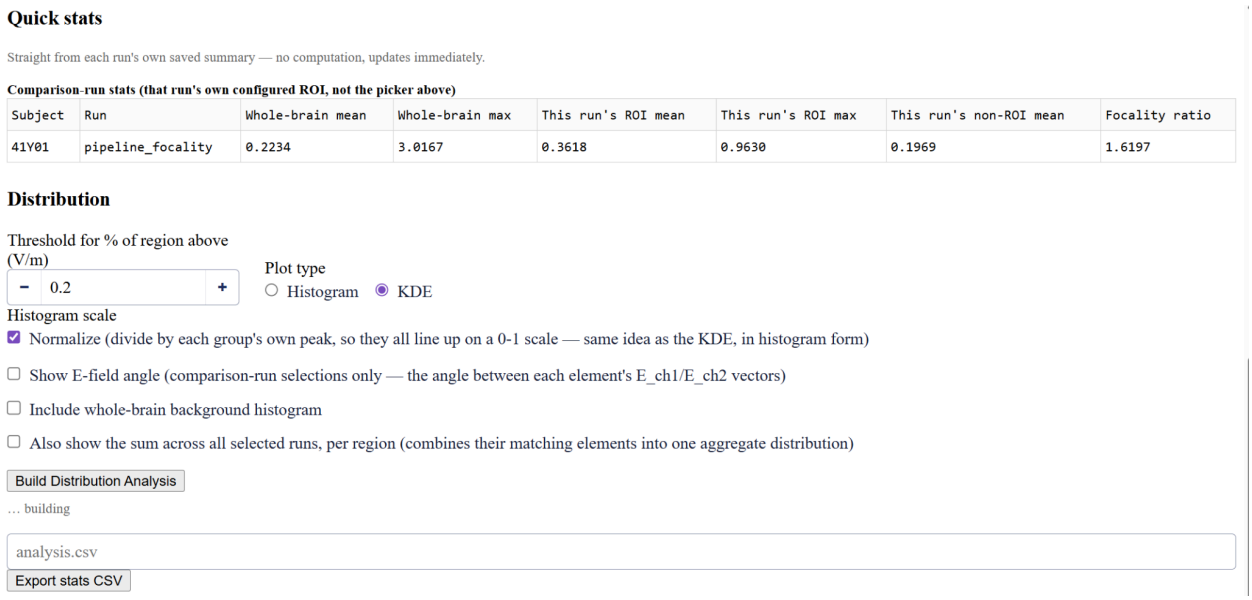
Every already-computed result for the selected subject(s) — pick one or more to analyze (several at once overlays them).

	Subject	Source	Run / label	E-field data
☐	41Y01	comparison	41Y01_row0	✓
☐	41Y01	comparison	manual_focality	✓
☐	41Y01	comparison	pipeline_focality	✓
☐	41Y01	comparison	setup	✓
☐	41Y01	comparison	template_focality	✓
☐	41Y01	TIoptimization	sub-41Y01_roi-striatum_ROI_goal-ex_focality_job-66213799_run-20260821_111126	–
☐	41Y01	TIoptimization	sub-41Y01_roi-striatum_ROI_goal-ex_focality_job-66501809_run-20260914_101846	–
☐	41Y01	TIoptimization	sub-41Y01_roi-striatum_ROI_goal-ex_focality_job-66503299_run-20260914_135009	–

Select all | Deselect all

This part of the pipeline assumes there are already E-field simulations present (or that the pipeline has been run and data has been synced from SCITAS).

4.1) Quick Stats and plots



Quick stats

Straight from each run's own saved summary — no computation, updates immediately.

Comparison-run stats (that run's own configured ROI, not the picker above)

Subject	Run	Whole-brain mean	Whole-brain max	This run's ROI mean	This run's ROI max	This run's non-ROI mean	Focality ratio
41Y01	pipeline_focality	0.2234	3.0167	0.3618	0.9630	0.1969	1.6197

Distribution

Threshold for % of region above (V/m)

− 0.2 +

Plot type

○ Histogram ◉ KDE

Histogram scale

☑ Normalize (divide by each group's own peak, so they all line up on a 0-1 scale — same idea as the KDE, in histogram form)

☐ Show E-field angle (comparison-run selections only — the angle between each element's E_ch1/E_ch2 vectors)

☐ Include whole-brain background histogram

☐ Also show the sum across all selected runs, per region (combines their matching elements into one aggregate distribution)

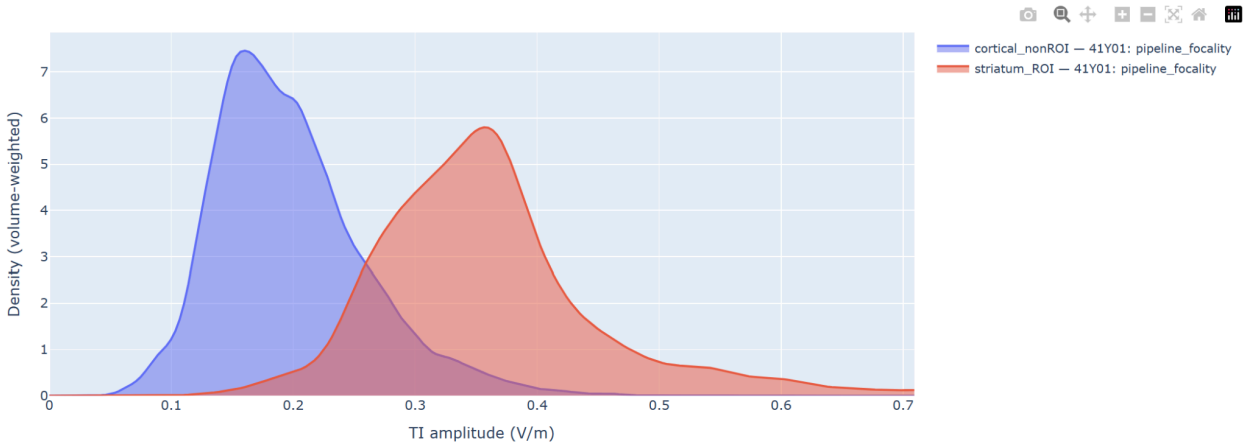
Build Distribution Analysis

… building

analysis.csv

Export stats CSV

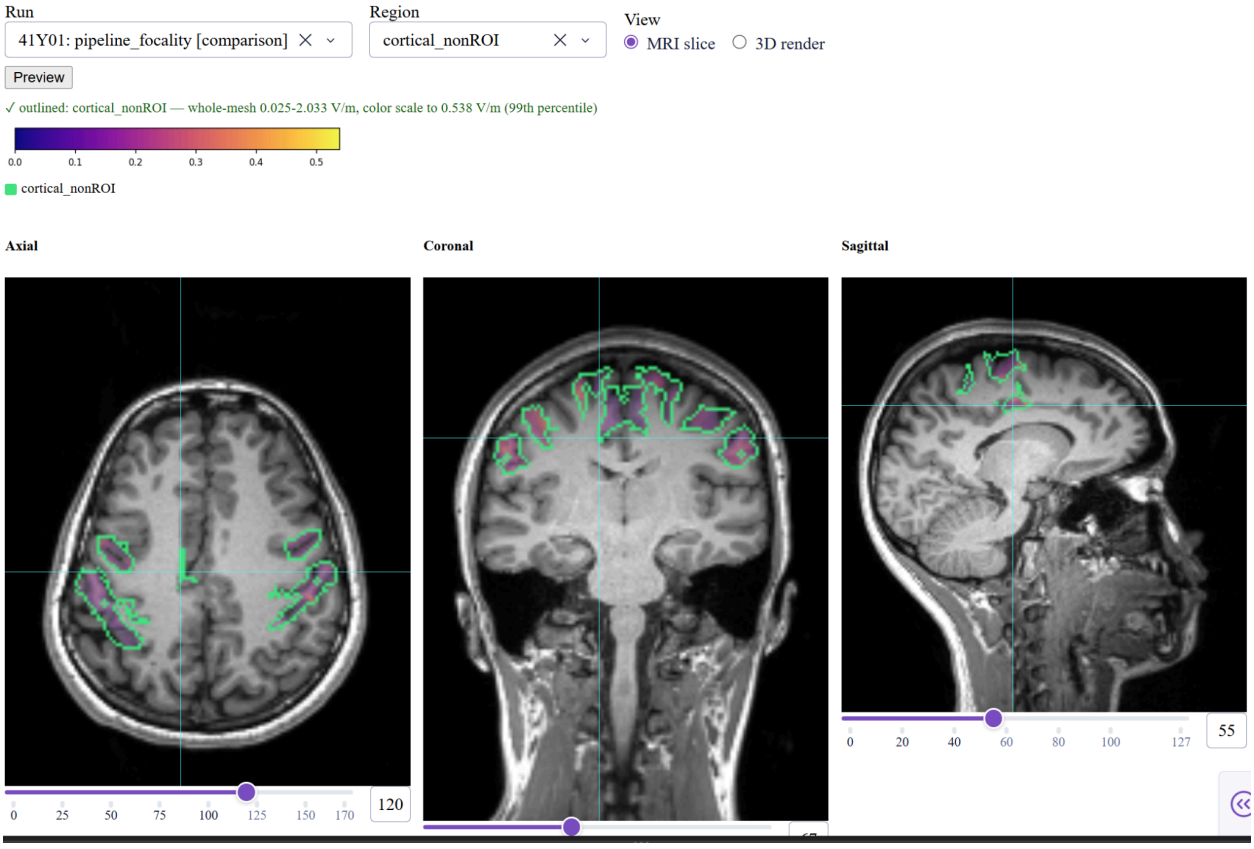
With this, you can be able to plot data about your runs, including the distribution of the E-fields, the angles if available, and information about each region.

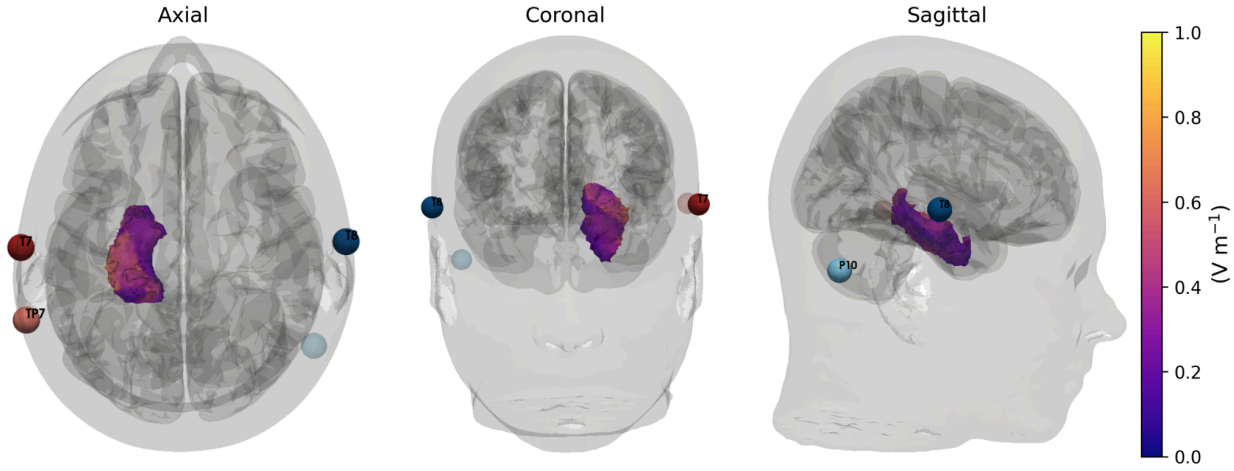


Region	Run	N elements	Mean	Median	p90	p95	p99	Max	% above threshold
cortical_nonROI	41Y01: pipeline_focality	92887	0.1972	0.1860	0.2703	0.3025	0.3796	0.5159	42.3%
striatum_ROI	41Y01: pipeline_focality	22736	0.3619	0.3572	0.5276	0.6151	0.7695	0.9630	98.3%

4.2) MRI and 3D viewer

This tool is here to help you scroll through your ROIs as in an MRI, and visualize using heatmaps the distribution of the E-field throughout your regions. You can also display the ROIs as a single 3D image for figures.
This is also available in the Simulation pages.




Axial
Coronal
Sagittal
T7
T8
TP7
P10
1.0
0.8
0.6
0.4
0.2
0.0
(V m^{-1})